SOLUTIONS: DESIGN OF LINEAR MULTIVARIABLE CONTROL SYSTEMS

1. a) Let $\Delta(s)$ and $H(s)$ be stable. The feedback loop is stable if $\|\Delta H\|_\infty < 1$.

b) Let the input to Δ_m be ε while the output be δ. Then a calculation shows that $\varepsilon = S\delta$ where $S = (I+GK)^{-1}$ is the sensitivity. Using the small gain theorem, to satisfy the robustness design requirement, it is sufficient that $\|\Delta_m(j\omega)S(j\omega)\| < 1, \forall\omega$. This can be satisfied if $\|W_m S\|_\infty < 1$, where $W_m = w_m I$.

A calculation shows that, when $\Delta_m = 0$,

$$u(s) = K(s)S(s)r(s), \qquad y(s) = G(s)K(s)S(s)r(s).$$

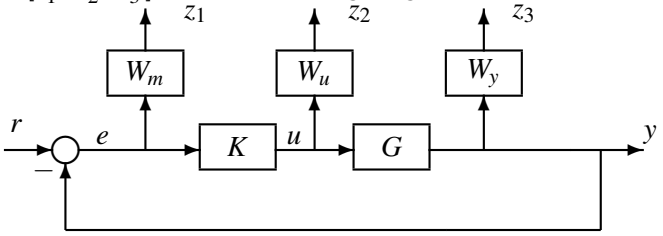
Thus

$$\|u(j\omega)\| \le \|K(j\omega)S(j\omega)\|\|r(j\omega)\|, \ \|y(j\omega)\| \le \|G(j\omega)K(j\omega)S(j\omega)\|\|r(j\omega)\|.$$

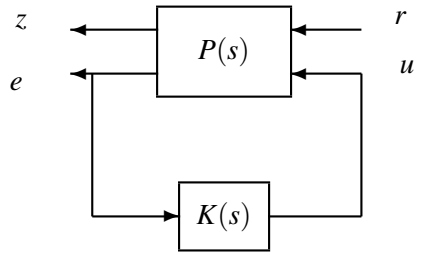
It follows that sufficient conditions to achieve the performance design specifications are $\|K(j\omega)S(j\omega)\| < |w_u^{-1}(j\omega)|$ and $\|G(j\omega)K(j\omega)S(j\omega)\| < |w_y^{-1}(j\omega)|$ for all ω or equivalently $\|W_u KS\|_\infty < 1$ and $\|W_y GKS\|_\infty < 1$, where $W_u = w_u I$ and $W_y = w_y I$.

To satisfy all design requirements, it is sufficient that $\left\| \begin{bmatrix} W_m S \\ W_u KS \\ W_y GKS \end{bmatrix} \right\|_\infty < 1.$

c) So, the design specifications reduce to the requirement that the transfer matrix from r to $z = [z_1^T \ z_2^T \ z_3^T]^T$ in the following diagram has $\mathscr{H}_\infty$-norm less than 1.



d) The generalized regulator problem: find an internally stabilizing K such that $\|T_{zr}\|_\infty < 1$



$$P = \begin{bmatrix} P_{11} & P_{12} \\ P_{21} & P_{22} \end{bmatrix} = \left[\begin{array}{c|c} W_m & -W_m G \\ 0 & W_u \\ 0 & W_y G \\ \hline I & -G \end{array}\right].$$

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2. a) It is assumed that the triple (A,B,C) is minimal. A suitable Lyapunov function for regulating x is $V = x^T Px$ where $P = P^T$. Set $u = -Fx$. Provided that $P = P^T \succ 0$ and $\dot{V} < 0$ along closed-loop trajectories, we can assume $\lim_{t\to\infty} x(t) = 0$. Then $\dot{V} = \dot{x}^T Px + x^T P\dot{x} = x^T \left(A^T P + PA - F^T B^T P - PBF\right) x$. Using $x(\infty) = 0$, $\int_0^\infty \dot{V}(t)dt = \int_0^\infty x^T \left(A^T P + PA - F^T B^T P - PBF\right) xdt = -x_0^T Px_0$.

b) Let $R = \alpha^2 I$. Adding the last equation to the expression for J and completing a square:

$$J = x_0^T Px_0 + \int_0^\infty \{x^T [A^T P + PA + C^T C - PBR^{-1}B^T P]x + \left\|\alpha(F - R^{-1}B^T P)x\right\|^2\} dt.$$

Since the last term is always nonnegative, it follows that the minimizing value of F is given by $F = R^{-1}B^T P$. We can set the term in square brackets to zero provided P satisfies the Riccati equation $A^T P + PA + C^T C - PBR^{-1}B^T P = 0$. It follows that the minimum value of J is $x_0^T Px_0$.

c) We prove that $A_c := A - BR^{-1}B'P$ is stable. The Riccati equation can be written as

$$A_c' P + PA_c + C^T C + PBR^{-1}B'P = 0.$$

Let $\lambda \in \mathscr{C}$ be an eigenvalue of A_c and $y \neq 0$ be the corresponding eigenvector. Pre– and post–multiplying the Riccati equation be y' and y respectively gives $(\lambda + \bar{\lambda})y'Py + y'C'Cy + y'PBR^{-1}B'Py = 0$. Since $P \succ 0$ and $y \neq 0$, $y'Py > 0$, $y'C'Cy \geq 0$ and $y'PBR^{-1}B'Py \geq 0$. It follows that $\lambda + \bar{\lambda} \leq 0$ with $\lambda + \bar{\lambda} = 0$ if and only if $B'Py = 0$ and $Cy = 0$. But this implies $Ay = \lambda y$ and $Cy = 0$, which contradicts the observability of the pair (A,C). This proves that $\lambda + \bar{\lambda} < 0$ and so the closed-loop is stable.

d) Now, $\dot{x} = Ax + Bu + Bd = Ax + B(d - Fx)$. It follows that

$$\begin{aligned} \dot{x} &= (A - BF)x + Bd \\ z_1 &= Cx \end{aligned}$$

it follows that $T_{z_1 d} \stackrel{s}{=} (A - BF, B, C, 0)$. It follows from the bounded real lemma that $||T_{z_1 d}||_\infty < \gamma$ if there exists $P = P^T \succ 0$ such that

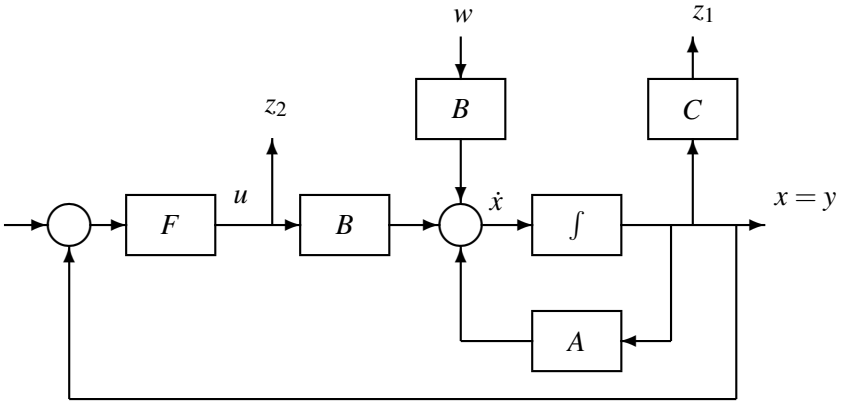
$$\begin{bmatrix} P(A - BF) + (A - BF)^T P + C^T C & PB \\ B^T P & -\gamma^2 I \end{bmatrix} \prec 0$$

Using a Schur complement argument, this inequality is equivalent to

$$P(A - BF) + (A - BF)^T P + C^T C + \gamma^{-2} PBB^T P \prec 0.$$

However, it follows from the Riccati equation in Part c above that $P(A - BF) + (A - BF)^T P + C^T C + \alpha^{-2} PBB^T P = 0$. Ignoring issues relating to strict matrix inequalities, this proves that $||T_{z_1 d}||_\infty \leq \alpha$ and so α is an upper bound on $||T_{z_1 d}||_\infty$.

3. a) The block diagram is given as:



b) Let $J = \|z\|_2^2 - \gamma^2\|w\|_2^2$ denote the cost function. Let $V = x'Xx$ and set $u = Fx$. Provided that $X = X' \succ 0$ and $\dot{V} < 0$ along the closed-loop trajectory, we can assume $\lim_{t\to\infty} x(t) = 0$. Then

$$\dot{V} = \dot{x}'Xx + x'X\dot{x} = x'\big(A'X + XA + F'B'X + XBF\big)x + x'XBw + w'B'Xx.$$

Integrating from 0 to ∞ and using $x(\infty) = 0$,

$$0 = \int_0^\infty [x'\big(A'X + XA + F'B'X + XBF\big)x + x'XBw + w'B'Xx]dt.$$

Using the definition of J, adding the last equation, and completing two squares,

$$J = \int_0^\infty \{x'\left[A'X + XA + C'C - (1-\gamma^{-2})XBB'X\right]x + \|Zx\|^2 - \|v\|^2\}dt$$

where $Z = F + B'X$ and $v = \gamma w - \gamma^{-1}B'Xx$.

c) The requirement $\|H\|_\infty \le \gamma$ is equivalent to $J \le 0$. Thus sufficient conditions for $J \le 0$ are $Z = 0$ and the existence of X such that

$$A'X + XA + C'C - (1-\gamma^{-2})XBB'X = 0, \qquad X = X' \succ 0.$$

Setting $Z = 0$, the control policy is $F = -B'X$. Setting $v = 0$, the worst case disturbance is $w^* = \gamma^{-2}B'Xx$. The closed-loop with $u = Fx$ and $w = w^*$ is $\dot{x} = A_c x$ where $A_c = A - (1-\gamma^{-2})BB'X$ and a further condition is $Re\ \lambda_i(A_c) < 0,\ \forall\ i$.

d) Using the solution in Part c) with $\gamma = 0$, it follows that the Riccati equation becomes

$$A'X + XA + C'C = 0, \qquad X = X' \succ 0.$$

Thus, since $A_c = A$, it follows that a condition for the solution of the problem is that A is stable.

4. a) Since A is stable, $\|H\|_\infty < \gamma$ if and only if, with $x(0)=0$, $J := \int_0^\infty [y^T y - \gamma^2 u^T u]dt < 0$, for all $u(t)$ such that $\|u\|_2 < \infty$. If $\|u\|_2$ is bounded, then $\lim_{t\to\infty} x(t) = 0$. Let $P = P^T$. Now, $\int_0^\infty \frac{\mathrm{d}}{\mathrm{dt}}[x^T Px]dt = x(\infty)^T Px(\infty) - x(0)^T Px(0) = 0$. So,

$$0 = \int_0^\infty \left(\dot{x}^T Px + x^T P\dot{x}\right) dt = \int_0^\infty [x^T(A^T P + PA)x + x^T PBu + u^T B^T Px]dt.$$

Use $y = Cx + Du$ and add the last expression to J

$$\begin{aligned} J &= \int_0^\infty [x^T(A^T P + PA + C^T C)x + 2x^T(PB + C^T D)u + u^T(D^T D - \gamma^2 I)u]dt \\ &= \int_0^\infty \begin{bmatrix} x^T & u^T \end{bmatrix} \overbrace{\begin{bmatrix} A^T P + PA + C^T C & PB + C^T D \\ B^T P + D^T C & D^T D - \gamma^2 I \end{bmatrix}}^{M} \begin{bmatrix} x \\ u \end{bmatrix} dt. \end{aligned}$$

It follows that $J < 0$, and so $\|H\|_\infty < \gamma$, if $P \succ 0$ and $M \prec 0$.

b) i) An inspection of Figure 4 shows that

$$\begin{aligned} \dot{x} - \dot{\hat{x}} &= (A + LC)(x - \hat{x}) + \begin{bmatrix} B & L \end{bmatrix} \begin{bmatrix} w_1 \\ w_2 \end{bmatrix} \\ z &= F(x - \hat{x}) \end{aligned}$$

It follows that

$$T_{zw}(s) \stackrel{s}{=} \left[\begin{array}{c|cc} A + LC & \begin{bmatrix} B & L \end{bmatrix} \\ \hline F & \begin{bmatrix} 0 & 0 \end{bmatrix} \end{array}\right] \stackrel{s}{=:} \left[\begin{array}{c|c} A_c & B_c \\ \hline C_c & D_c \end{array}\right]$$

ii) The Bounded Real Lemma states that A_c is stable $\|T_{zw}\|_\infty < \gamma$ if there exists a $P = P^T$ such that

$$\begin{aligned} \begin{bmatrix} A_c^T P + PA_c + C_c^T C_c & PB_c + C_c^T D_c \\ B_c^T P + D_c^T C_c & D_c^T D_c - \gamma^2 I \end{bmatrix} &\prec 0 \\ P = P^T &\succ 0 \end{aligned}$$

By substituting the expressions for A_c, B_c, C_c and D_c, this becomes

$$\begin{aligned} \begin{bmatrix} (A + LC)^T P + P(A + LC) + F^T F & PB & PL \\ \star & -\gamma^2 I & 0 \\ \star & \star & -\gamma^2 I \end{bmatrix} &\prec 0 \\ P = P^T &\succ 0 \end{aligned}$$

where $\star$ denotes terms easily inferred from symmetry.

iii) By defining $Y = PL$, the matrix inequalities become

$$\begin{aligned} \begin{bmatrix} PA + A^T P + YC + C^T Y^T + F^T F & PB & Y \\ \star & -\gamma^2 I & 0 \\ \star & \star & -\gamma^2 I \end{bmatrix} &\prec 0 \\ P = P^T &\succ 0 \end{aligned}$$

which are linear.